

# Requirement Specification Document

## E-Commerce Order Management Database System

*Week 1 – Requirement Analysis and Customer Database Module*

|                 |                                        |
|-----------------|----------------------------------------|
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| Project Title   | Real-Time Attendance Management System |
| Week            | Week 1                                 |

## 1. Introduction

### 1.1 Purpose

The purpose of this document is to analyze and specify the business requirements of the Real-Time Attendance Management System. The system is designed to record and monitor student attendance in real time, helping educational institutions maintain accurate attendance records.

### 1.2 Project Scope

The system allows administrators, faculty members, and students to manage attendance records. Faculty can mark attendance instantly, students can view attendance reports, and administrators can manage users, courses, and attendance history. The database ensures secure and reliable storage of attendance information.

### 1.3 Intended Users

- Students – View attendance records.
- Faculty – Mark and update attendance.
- Administrator-Manage students, faculty, courses, and reports.

## 2. Business Requirements

The system must:

- Store student information.
- Maintain faculty records.
- Record attendance in real time.
- Generate attendance reports.
- Calculate attendance percentage.
- Provide secure login for all users.

## 3. Functional Requirements

Functional requirements describe what the system must do.

| ID   | Requirement Description                                            |
|------|--------------------------------------------------------------------|
| FR-1 | The system shall allow students to register with personal details. |
| FR-2 | The system shall allow faculty to mark attendance.                 |
| FR-3 | The system shall store attendance date and time automatically.     |
| FR-4 | The system shall calculate attendance percentage.                  |
| FR-5 | The system shall generate daily and monthly attendance reports.    |

|      |                                                                   |
|------|-------------------------------------------------------------------|
| FR-6 | The system shall allow administrators to manage student records   |
| FR-7 | The system shall allow students to view their attendance history. |
| FR-8 | The system shall send attendance notifications when required.     |

## 4. Non-Functional Requirements

- **Performance:** Attendance should be recorded within 2 seconds.
- **Security:** User data must be securely stored.
- **Reliability:** Attendance records must not be duplicated.
- **Usability:** The interface should be simple and user-friendly.
- **Scalability:** The system should support thousands of students.

## 5. Customer Database Module – Data Requirements

Since this week's focus includes the Customer Database Module, the following entity and attributes have been identified:

| Attribute             | Data Type         | Description           |
|-----------------------|-------------------|-----------------------|
| student_id            | INT (Primary Key) | Unique Student ID     |
| student_name          | VARCHAR(100)      | Student Name          |
| department            | VARCHAR(50)       | Department            |
| phone_number          | VARCHAR(10)       | Contact Number        |
| attendance_percentage | VARCHAR(5,2)      | Attendance Percentage |
| registration_date     | DATE              | Registration Date     |

## 6. Assumptions and Constraints

- The system will use a relational database such as MySQL or Oracle.
- Only registered students can have attendance recorded.
- Every attendance record belongs to one student.
- This document covers Week 1 requirements only. Future modules such as Faculty, Course, and Attendance Reports will be developed in later weeks.

## 7. Conclusion

This document defines the initial business and functional requirements of the Real-Time Attendance Management System, focusing on the Student Database Module. It serves as the foundation for database design, ER diagrams, normalization, and future implementation.